Few-Shot Voice Cloning for Vocal Preservation

What Helps More in F5-TTS, Reference Audio or LoRA?

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Abstract

This project studies a simple question with real practical value: if we want to preserve a person's voice with very little data, what matters more in F5-TTS, giving the model a short reference clip at generation time or fine-tuning it with LoRA? We evaluated 45 settings by varying reference audio length from 0 to 8 seconds and LoRA training data from 0 to all remaining training sentences. In total, we generated 675 samples and scored them with speaker similarity and word error rate.

In this setup, the answer is clear. Reference audio matters far more than LoRA. Moving from 0 to 1.5 seconds of reference audio improves speaker similarity by about +0.34, while the best LoRA gain at a fixed anchor is only +0.025. A larger LoRA adapter helps a little, but not enough to replace reference audio. The main practical lesson is simple: for voice preservation with F5-TTS, a short clean reference recording is the key ingredient. Fine-tuning is optional and gives only small returns in this single-speaker study.

If you can collect one clean reference clip, do that first. In our experiments, the reference clip did most of the work. LoRA changed the results only a little, even when we gave it all available training sentences and even when we increased the adapter size.

Introduction

People with ALS, Parkinson's, and related conditions can gradually lose the ability to speak in their own voice. Standard text-to-speech systems can still say the words, but they do not preserve identity. That is the real motivation here. The goal is not only intelligible speech. It is speech that still sounds like the person.

The original project proposal framed the task exactly this way: build a voice cloning pipeline that needs very little input data and can help preserve a person's vocal identity.

F5-TTS can adapt to a speaker in two different ways:

1. **Reference audio at generation time.** The model hears a short clip of the target speaker right before it generates speech.

2. **LoRA fine-tuning.** A small set of trainable weights is updated on that speaker's recordings.

The project question is: *when does LoRA actually add value beyond the reference clip?*

How F5-TTS Works Under the Hood

Before diving into the experiment, it helps to understand what F5-TTS actually does inside. This section gives a simplified walkthrough of the model's architecture and its training method, Flow Matching. The goal is to build enough intuition so the experimental results make sense.

The Big Picture

F5-TTS turns text into speech in two main stages. First, a neural network called a Diffusion Transformer (DiT) generates a mel-spectrogram, which is a visual representation of sound that captures frequency content over time. Second, a separate model called a vocoder (specifically, Vocos) converts that spectrogram into an actual audio waveform you can listen to. The heavy lifting happens in the first stage. The vocoder is pretrained and stays frozen throughout.

The model takes three inputs at generation time: (1) the text you want it to say, (2) a short reference audio clip of the target speaker, and (3) the transcript of that reference clip. The reference audio gets converted into a mel-spectrogram and concatenated with noise along the time axis. The text (both the reference transcript and the target sentence) is converted into a sequence of character tokens. The DiT then processes both streams together and learns to predict a clean spectrogram that continues the reference audio with the new sentence, in the same voice.

This design is what makes zero-shot voice cloning possible. The model does not need to be retrained for a new speaker. It just needs to "hear" a few seconds of that speaker's voice as context, and it can extend the audio with new words in the same style. This is similar in spirit to how a large language model can adapt to a writing style from a prompt, except here the "prompt" is audio.

The Diffusion Transformer

The core of F5-TTS is a transformer with about 339.6 million parameters. Unlike a standard language transformer that predicts the next token in a sequence, this one predicts a velocity field that gradually transforms noise into a mel-spectrogram. Each transformer block contains self-attention layers (which let different parts of the spectrogram and text attend to each other) and feed-forward layers (which process each position independently). The text and audio representations are combined inside the model through a shared attention mechanism, so the transformer can learn the alignment between characters and sounds without needing a separate alignment module.

This is worth highlighting: many older TTS systems require an explicit duration predictor or attention alignment step to figure out which part of the text maps to which part of the audio. F5-TTS skips that entirely. The paper calls this a "non-autoregressive" design with "no explicit duration or alignment modeling." The transformer figures out timing on its own, guided by the reference audio and the flow matching objective.

What Is Flow Matching?

Flow Matching is the training method that teaches the DiT how to generate spectrograms. The core idea is surprisingly simple.

Imagine you have a clean spectrogram (from real speech) and a blob of pure random noise, both the same shape. Flow Matching defines a straight path between the noise and the clean spectrogram. At any point along that path, say 30% of the way from noise to clean, you get a partially noisy spectrogram. The model's job during training is simple: given that partially noisy spectrogram and the current position on the path (a number between 0 and 1, called the timestep), predict the direction and speed you need to travel to reach the clean spectrogram. In mathematical terms, the model learns a velocity field.

More concretely, during each training step: (1) pick a real spectrogram from the dataset, (2) sample a random timestep t between 0 and 1, (3) create a noisy version by blending the clean spectrogram with Gaussian noise according to t, and (4) train the model to predict the velocity that would move from the current noisy state toward the clean target. The loss function is just the mean squared error between the predicted velocity and the true velocity along the straight path.

This is closely related to diffusion models, but cleaner in some ways. Traditional diffusion models define a complex noising schedule and learn to reverse it step by step. Flow Matching instead uses straight-line paths (called optimal transport paths), which tend to be easier to learn and faster to sample from. The F5-TTS paper uses the term Conditional Flow Matching (CFM) because the flow is always conditioned on the text and reference audio.

How Inference Works

At inference time, the model starts from pure noise and follows the learned velocity field step by step until it arrives at a clean spectrogram. This is done using an ODE (ordinary differential equation) solver. You can think of it like navigating by compass: at each step, the model asks "which direction should I go from here?" and takes a small step in that direction. After enough steps, the noise has been transformed into a realistic spectrogram.

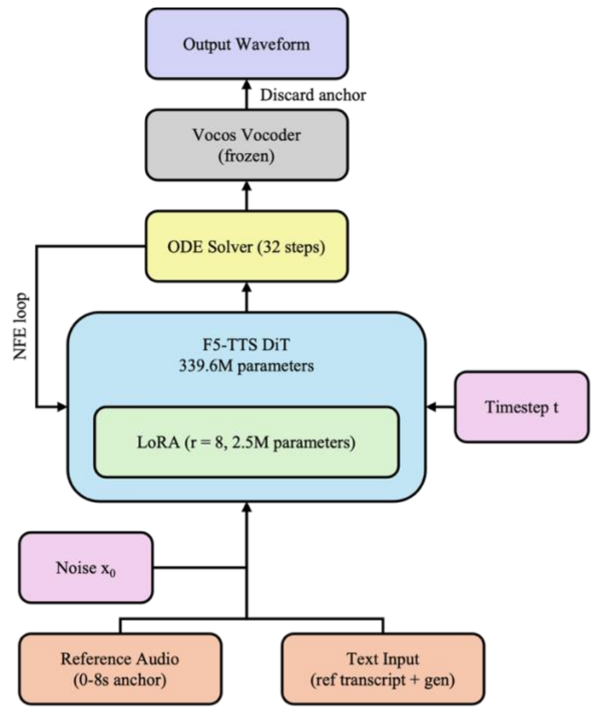
In this project, we used 32 steps (called NFE, for number of function evaluations). More steps generally give higher quality but take longer. The model also uses classifier-free guidance (CFG), which is a technique to make the output more closely follow the conditioning signals (the text and reference audio) at the cost of slightly less diversity.

One important detail for this project: the model estimates the total output length based on the ratio between the reference audio length and the reference text length. If the reference clip is 3 seconds long and covers 20 characters of text, and the new sentence has 40 characters, the model expects to generate roughly 6 seconds of new audio. This is exactly the mechanism that breaks down at the 1.5-second anchor setting, as discussed later in the results.

Where LoRA Fits In

LoRA (Low-Rank Adaptation) is applied to the attention and feed-forward layers inside the DiT. Specifically, it targets the query, key, value, and output projections in each attention block, plus two layers in each feed-forward block. Instead of updating the full weight matrices during fine-tuning (which would mean moving all 339.6 million parameters), LoRA freezes the original weights and adds a small trainable bypass. This bypass is a pair of small matrices that together form a low-rank update. With rank 8, the default setting in this project, that adds about 2.52 million trainable parameters, roughly 0.74% of the total model.

The idea is that LoRA could encode speaker-specific patterns into these small weight updates, so the model becomes slightly specialized for one voice. The experiment tests whether that specialization actually helps compared to just giving the model a good reference clip at generation time.



Simplified Inference Pipeline Diagram.

Dataset and Split

We recorded one speaker reading 20 Harvard sentences. These sentences are commonly used in speech work because they cover a broad set of English sounds. The recording lasted about five minutes.

Stable Whisper was used for forced alignment, which gave us 20 sentence-level clips and 159 word-level clips. The setup was:

- **Reference anchor:** the first 3 sentences, about 7.2 seconds in total.
- **LoRA training pool:** all remaining sentence clips.
- **Test set:** 5 new sentences that were not spoken in the recording.

A small but important detail: the notebook requested LoRA training sizes of 0, 1, 5, 10, and 20 sentences, but after reserving 3 sentences for the anchor, only 17 training sentences were actually available. So the last column in this revised report is labeled **all remaining (17)**. The numerical results are unchanged. They just needed a more accurate label.

Metrics

We used two automated metrics.

- **Speaker similarity** (higher is better, roughly 0 to 1). We used Resemblyzer to embed both the generated audio and the target speaker audio, then measured cosine similarity.
- **Word error rate, WER** (lower is better). We transcribed the generated speech with Whisper and compared that transcription to the target sentence.

These metrics do not cover everything. They say a lot about identity matching and intelligibility, but they do not directly measure naturalness or emotional quality. Still, they are strong first-pass diagnostics for this question.

Experimental Setup

We ran a full grid over two knobs:

- **Reference audio at generation:** 0, 0.5, 1, 1.5, 2, 3, 4, 6, and 8 seconds.
- **LoRA training size:** 0, 1, 5, 10, and all remaining 17 sentences.

This gives 9 x 5 = 45 conditions. For each condition, we generated all 5 test sentences with 3 random seeds, for 15 samples per condition and **675 samples total**.

We used pretrained F5-TTS v1 with 32 flow-matching steps at inference. LoRA used rank $r = 8$, scaling $\alpha = 16$, dropout 5%, AdamW, cosine decay, and gradient accumulation of 4. We also ran two follow-up analyses:

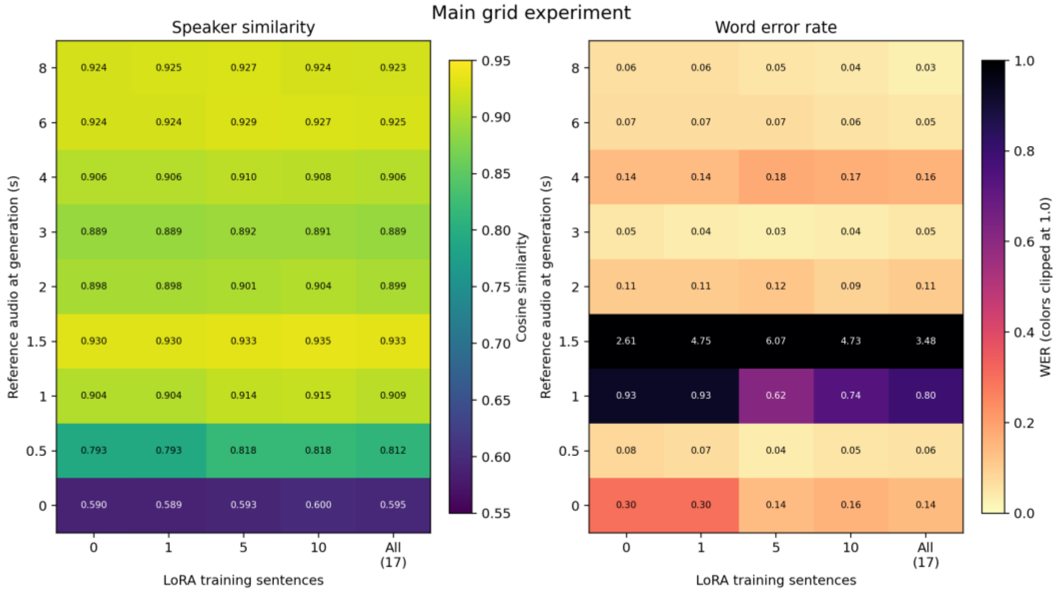
1. **Anchor diagnostics,** to understand the strange behavior around 1.5 to 3 seconds.
2. **LoRA rank sweep,** with ranks 2, 4, 8, 16, and 32.

All runs were done on an NVIDIA H100 80GB GPU in Google Colab.

Results

The Main Result: Reference Audio Dominates

The full grid is easy to read: moving down the rows changes the numbers a lot, while moving across the columns changes them only a little. In other words, reference audio has a large effect and LoRA has a small one.

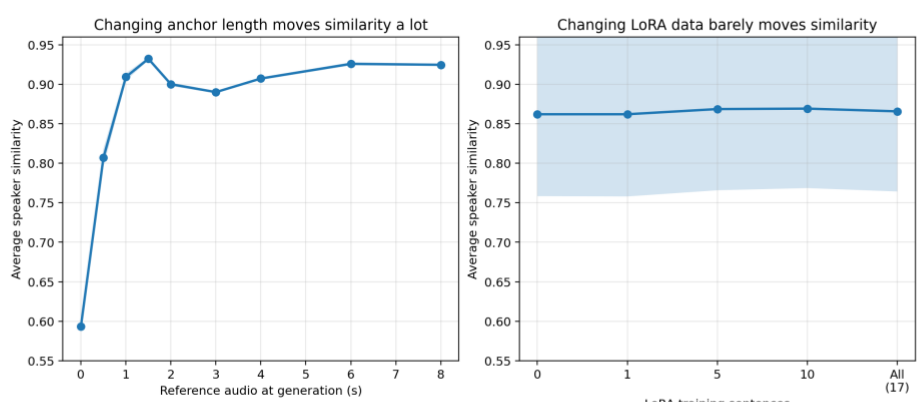


Main grid experiment. Left: speaker similarity. Right: WER. The last column is relabeled to "all remaining (17)" because the dataset contained 17 non-anchor training sentences. The strongest changes happen across anchor durations, not across LoRA training sizes.

The quantitative summary is even clearer:

- Going from 0 to 1.5 seconds of reference audio, with no fine-tuning, increases similarity from 0.590 to 0.930. That is a gain of **+0.340**.
- The best LoRA improvement at a fixed anchor is only **+0.025**, from 0.793 to 0.818 at the 0.5 second anchor.
- At 0 seconds of reference audio, where LoRA should have the best chance to help, the improvement from no training to all remaining training sentences is only **+0.005**. That is smaller than the standard deviation.

Averaging across the whole grid tells the same story. Average similarity changes sharply with anchor length, but it is nearly flat across LoRA training size.



Average speaker similarity when changing one factor at a time. Left: anchor duration strongly changes the outcome. Right: LoRA training size barely changes the average result.

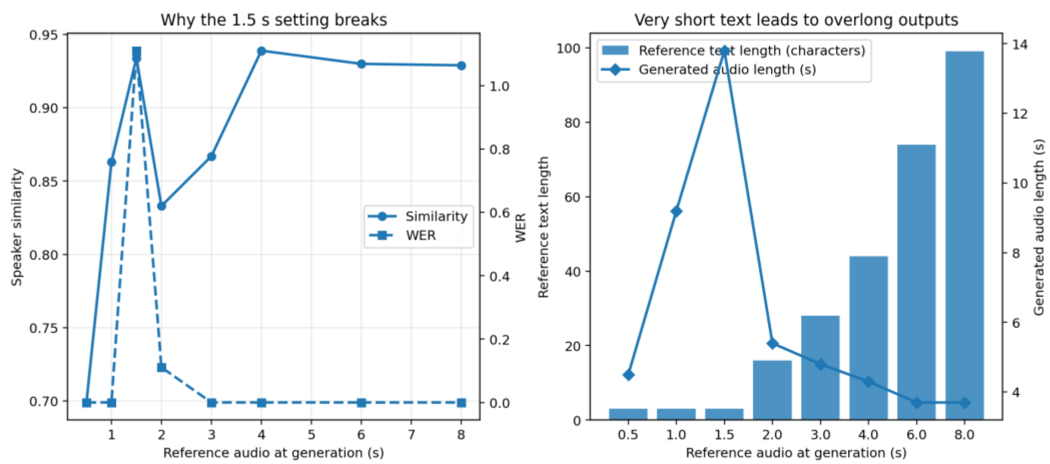
Speech Accuracy Mostly Stays Good, Except for a Failure Region

Most settings have low WER. The cleanest region is roughly 3 to 8 seconds of reference audio, where WER is usually below 0.07. The one obvious failure region is the 1.5 second anchor, where WER jumps far above 1.0 and sometimes above 6.0.

That is not a small degradation. It means the model often repeats the sentence or overgenerates badly. This matters because the highest similarity scores in the grid happen exactly in that broken region. So similarity alone would tell the wrong story.

Why the 1.5 Second Setting Breaks

The diagnostic experiment explains the failure. Around 1.5 seconds, the reference audio is already long enough to carry strong speaker cues, but the transcript paired with that truncated audio is still extremely short. In our case, it is effectively just “The”. F5-TTS uses both reference audio length and reference text length when it estimates how long the generated utterance should be. That mismatch causes the model to allocate too much output time, and it fills that time by repeating itself.



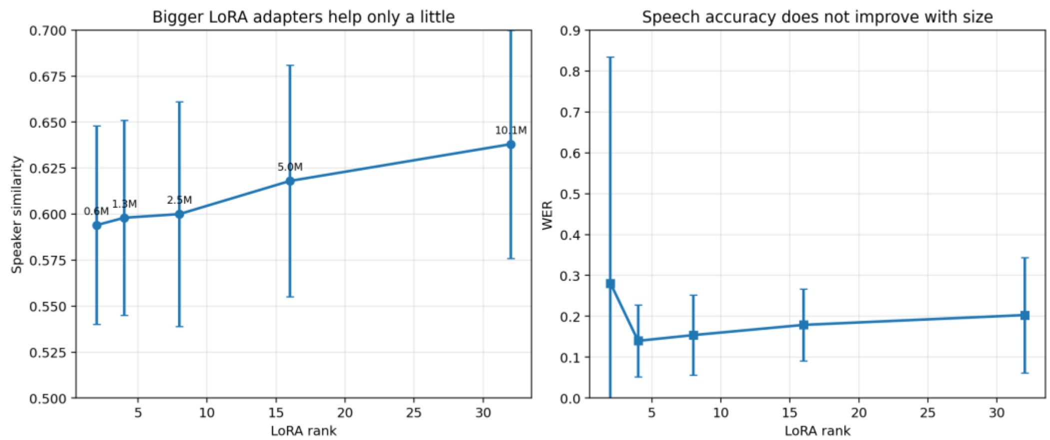
Anchor-duration diagnostics with no LoRA. Left: similarity rises quickly, but WER spikes at 1.5 seconds. Right: the spike is explained by a mismatch between very short reference text and much longer reference audio, which leads to overlong generated outputs.

There is also a weaker dip around 2 to 3 seconds. The likely reason is that the reference transcript is still a sentence fragment there, which makes text-audio alignment less stable. Once the anchor reaches a complete sentence, performance becomes more reliable.

The practical lesson is not that 1.5 seconds is bad in general. It is that **the audio-text pair used as reference must stay well aligned**. A short but clean reference clip can work well, but a badly truncated one can fail badly.

Bigger LoRA Adapters Do Not Solve the Problem

We also tested whether a much larger LoRA adapter could make up for missing reference audio. This was done in the hardest setting: 0 seconds of reference audio, with all remaining training sentences.



LoRA rank sweep at 0 seconds of reference audio. Increasing rank from 2 to 32 raises similarity only modestly, from 0.594 to 0.638, and does not improve WER.

The effect is real but small. Increasing rank from 2 to 32 increases similarity by about +0.044, while the adapter size grows by about 16x. That is a poor trade. More importantly, even the largest adapter does not get close to what 1 second of reference audio already achieves without any fine-tuning.

Training Behavior

Training loss was noisy across all LoRA runs, which is not unusual for flow-matching models. More important, the checkpoint with the lowest loss was not reliably the one with the best speaker similarity. Choosing checkpoints by validation similarity instead of training loss was the right call.

That point is easy to miss, but it matters for future work. If the target is speaker preservation, then the model should be selected with a speaker-preservation metric, not only a generic optimization loss.

Discussion

What the Results Mean

The central conclusion is simple: in this single-speaker study, **reference audio is the main source of speaker identity**. LoRA helps only a little.

That does not mean LoRA is useless. It may still help with other models, larger speaker datasets, or more varied recording conditions. But in this project, it was not the thing driving performance.

A more careful version of the conclusion is:

- If the goal is to sound like the target speaker, collect a short clean reference clip first.
- If the goal is to keep the system simple and practical, zero-shot F5-TTS is already strong.
- If you do use LoRA, expect incremental gains, not a substitute for reference audio.

Practical Recommendations

Based on the results, the strongest recommendations are:

1. **Prioritize a clean reference clip.** That is the highest-return data you can collect.
2. **Use a complete and correctly aligned reference transcript.** The 1.5 second failure shows how costly misalignment can be.
3. **Treat 3 to 8 seconds as a useful operating region in this setup.** Around 3 seconds, WER is very low. Around 6 to 8 seconds, similarity is strongest while WER stays low.
4. **Do not expect larger LoRA ranks to rescue a no-reference setup.** They help only a little and can hurt clarity.

Limitations

The study has clear limits.

- **One speaker only.** We do not know yet how stable the pattern is across voices, accents, ages, or disease-affected speech.
- **Same-session data.** Training clips and reference clips came from the same recording session, which is easier than real deployment.
- **Automated metrics only.** No human listening test was run.
- **One architecture only.** The conclusion applies to F5-TTS here, not automatically to every TTS system.

The result is convincing for this experiment, but not universal.

Conclusion

This project asked a practical question: in few-shot voice cloning with F5-TTS, what matters more, a short reference clip or LoRA fine-tuning? In this study, the answer is the reference clip, by a wide margin.

A few seconds of reference audio gave a large jump in speaker similarity. LoRA gave only small gains, even with all available training data and even when the adapter size was increased substantially. The cleanest practical strategy is therefore simple: use F5-TTS with a short, well-transcribed reference clip, and treat LoRA as optional.

That is good news for vocal preservation. It means a useful system may not need a long recording session or speaker-specific retraining. A short clean sample may already be enough to keep much of a person's vocal identity.

References

1. Chen, Y., Chen, Z., et al. *F5-TTS: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching.* arXiv preprint arXiv:2410.06885, 2024.